

Testing a Venus flytraps's memory

Venus flytrap glows green in response to outside stimulus, providing insight into its memory. <http://digit.in/nov20-25>



Even bats know better

Wild vampire bats, upon getting sick, follow social distancing protocols. <http://digit.in/nov20-26>.

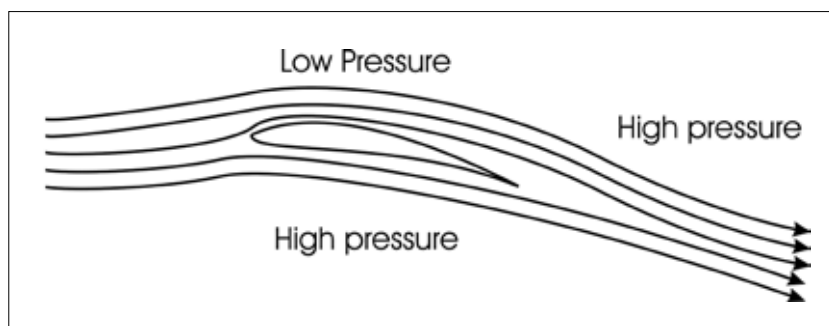
aerofoil means that the air molecules on the top edge of the wing move faster than the air molecules on the bottom edge of the wing, in order to cover the same distance. This means that the air on top of the wing moves at a high velocity and low pressure, and the air on the bottom moves at a low velocity and high pressure, producing an upward lift. The theory incorrectly assumes that the air molecules towards both the sides of the wing move at equal speeds, reaching the end of the wings at the same time. Experiments have shown that the molecules on the top of the wing move faster than the molecules along the bottom. Also, aeroplanes with wings that are of the same length along the top and bottom sides also can fly. According to the skipping stone theory, the air molecules bouncing off the lower side of the wings provide it with the lift necessary. This is similar to a flat rock thrown at a low angle on the surface of the lake, hence the name. As the upper surface does not factor into the theory at all, one would expect that all wings that have the same shape on the bottom side will fly in the same manner. This is obviously not true. Also, the theory cannot explain spoilers, movable portions located on the upper surfaces of wings, which can adjust the amount of lift provided. Another is called the Venturi theory, which says that the top surface of a wing behaves like a Venturi nozzle. Imagine a pipe with a narrow constriction in the middle. According to the Venturi theory, the fluid moving through the nozzle increases in velocity and reduces in pressure. After passing through the constriction, the pressure returns to the ambient levels within the pipe. According to the Venturi theory, the top surface of a wing behaves like a nozzle, increasing the velocity of the air and reducing the pressure. The problem is, there is no second surface to produce a nozzle, and the wing of a plane is just not a nozzle. Also, just as the Skipping Stone theory discounts the top surface, the Venturi theory

discounts the bottom surface of the plane. This means that a box that has the same curvature on the top as a wing should be able to produce the same amount of lift.

Then we have the two main theories, both of which are correct, but do not explain the business of flight completely. Both these theories are named after the scientists that demonstrated the underlying principles, Bernoulli and Newton, though neither applied the theory specifically to the problem of flight. Bernoulli's theory is the more popular of the two, and according to it the lift is produced by the air molecules moving faster over the top surface rather than the

lously simple explanation, one that applies equally to cambered and symmetrical wings, as well as planes flying upside down. The problem is that even this theory does not explain the low pressure area along the top surface of the wing.

The thing here is that both Newton's theory and Bernoulli's theory for lift are correct, at least partially correct. Even the three "wrong" theories mentioned earlier have some aspects that get it right. Researchers are using modern approaches including computational fluid dynamics (CFD) along with some complex maths that involves Navier-Stokes Equations and Euler Equations to come up with



Airflow around an aerofoil

Credit: James M Glowria/Stanford

bottom. Note that the air molecules along both surfaces do not take the same time to travel over the distance of the wing. The problem with the theory is that it does not explain why a higher velocity of a wind means that the pressure is lowered. It also cannot explain how aircraft with wings where both the top and bottom surfaces are symmetrical, or planes with flat wings can stay in the air. A wing that has a larger surface area along the top is called cambered, while a wing with equal surface areas along both sides is called symmetrical. Planes with symmetrical or cambered wings can both fly upside down, something that the Bernoulli theory of lift also cannot explain.

The other main theory is simply Newton's third law. The wing of a plane pushes the air downwards, which then pushes the plane upwards. This is actually a marve-

a comprehensive theory of lift that does not have any holes in it. Such a theory would have to include both Newton and Bernoulli's theories, and explain the low pressure zone created along the top surface of the wing.

In fact this is such a tough problem that even Albert Einstein could not figure out how planes fly. He even went on to design a wing based only on theory, based on Bernoulli's Theorem and experiments showed that most conventional wing designs were much better than Einstein's design. Still, a German WWI fighter plane was modified according to Einstein's design, and flown by Paul Ehrhardt. The flight was such a disaster that the pilot was elated when he managed to land the plane without incident. Einstein wrote to Ehrhardt accepting his failure, noting, "That is what can happen to a man who thinks a lot, but reads little." 